

Behavior 4th Lec.

• Management of pregnant cow

3. Milking: 2-3 times a day by: hand milking
machine milking

Rules for good milking:

قواعد الحلب

- ① Clean milking or care to be taking before milking
for animal → Should clean flank region and udder
for milker → Should wash his hands
for milking Troughs → the milk pail should be clean
for milking place → Should be clean

② Gentle: Kind Treatment and calling her by Pet name and we should take care not to injure the teats by

① Rough handling ② Exerting extreme pressure

③ Continued stripping after the udder appears to be empty

③ Quick: depend on the secretion of oxytocin hormone "last only for 5 min"

④ Quiet: Should avoid noise, so it doesn't release adrenaline.

عشان هتفرش أدرينالين

بيفر كذا عشان البقرة تفرز اللبن لازم العضلات اللي حوالي الغدة البائية تنقبض فأخذ حلب البقرة أو

ان ايها يرفع عنها الـ Teats بتنتش (50) العصب اللي فيها بيبيات اشارات للفرع عشان يفرز هرمون الـ oxytocin يروح الـ smooth muscle ويقلها تنقبض فالدور كله فالهرمون دا (الو تاثير

بيفضل (5) دقائق بس! عشان كذا عملية الحلب لازم تكون سريعة

واله كان لازم يفرز هورموني عشان هتفرش noradrenaline or Adrenaline اللي بيمنحو افراز اللبن

⑤ Regular: Cow should be milked regularly 2 or 3 times a day of the time between milking should be divided into periods as nearly equal

مثلا لو هتفرش بيقر العروق 12 ساعة لو ثلاثة يبتا الفتر لا ساعات ومكنا

⑥ Complete: the milking should be complete, if milking is in complete there is a tendency to dry the cow

Caring for milk

drying off: stop milking of cow at least 2 months of pregnancy

Milk is kept Clean

Should be cooled

Drying off

- ① **Incomplete milking**: don't extract all the milk from the udder at milking time for first few days after the drying off period has begun. Milking is stopped completely.
لا يتم حلب البقرة كلها لوجبات يتم حلب بها لتر الحليب ٢٠ وبعدين ١٠ and soon
- ② **intermittent milking**: will be milked once a day for awhile then once in every next day if finally milking will be stopped all together.
ممكن لو يتم الحلب مرة في اليوم فاليوم بيقوا مرة وبعدين تبقى يوم اتلبها ديوم ٤ وهذا العدد ما نوقف
- ③ **Complete cessation**: producing 10 liters of milking per day, the udder fills until pressure increases enough to stop secretion inside the udder, the milk gradually re-absorbed from the gland until it becomes completely dry.
في حالة ان البقرة بتنتج ١٠ لتر خاليوم او اقل من قبل الحلبها نالعلان كذا بتنتج كمية صغيرة

Parturition: 2 week before the expecting date of Calving they should be Isolated from the herd to the clean & disinfected Calving shed of Maturity قبل الولادة بأسبوعين لازم البقوا الى حوزة تتنزل عن القطيع وتروح أوضة الولادة وتكون نظيفة ومعقمة

علامات اقتراب الولادة

Signs of approaching Parturition:

- ① Relaxation of Sacrospinale Ligaments. ترفخ عضلات السججة
- ② Slackening of the tissue of the Perineum and Vulva انخلاع الشئ والجلات
- ③ Distension of the udder & teats اغزارب مخاطية
- ④ Thelous discharge from the Vulva
- ⑤ The normally lower body temperature of cattle during afternoon below 39°C

Care of Pregnant Cow.

- ① Separate the pregnant cow in Calving Pen, clean, roomy - well ventilated & bedded & finally disinfected ^{مطهر} يتم عزلها في حوض لولادة ولازم يكون رفيفة، فضيفة، هتطفيه بالفرش
- ② Choose the site of Parturition
 - (a) Best site for Calving is the small Pasture free from woods or foreign bodies or wet areas افضل موقع للولادة اطراى الخالية من الاشجار والاعشاب الغريبة، والمكان الرطبة
 - (b) During period of adverse cold & wet weather → dry warm building with heat lamps خراب الجوابار او رطب من دافئ جاف مزود بالحرارة
- ✗ The disadvantages of Parturition is that the frequency of close observation of animal ^{تتبع المرافاة اسبغس هتخوف نتابع قريب مع المرافاة}

Parturient and Maternal behavior:

① Parturient Behavior

- (a) Signs of 1st stage of Parturition
- (b) Birth stage
- (c) Post - Parturient stage

(a) Sign of 1st Stage of Parturition

1. Loss of appetite & rumination فقد الشهية و قلة الرضاعة
2. Restlessness & preference of isolation القلق و تفضيل العزلة
3. Lying down and getting up again.
4. Kicking at abdomen & Licking around to flank تضرب نفسها في بطنها و يتقلب على الفلانك
5. frequent shifting of her position & passing of small amount of feces & urine تغير مرات متكررة في موضعها و اخراج كمية قليلة من البراز و البول
6. Regular straining indicated by ^{تنقوس} arching of the back & elevated tail (1-3 hrs) before the chorioallantoic membrane rupture اجاء منتظم من خلال تنقوس الظهر و الذيل يكون مرتفع لمدة 1-3 ساعات قبل تمزيق الغشاء المشيمي

(b) Birth Stage (expulsive stage)

1. This stage begins with rupture of chorioallantoic membrane تمزق الغشاء المشيمي
2. 95% of all Cows were recumbent at the actual time of delivery where there is a powerful straining at short interval. 95% من جميع الابقار كان (recumbent) في وقت الولادة بسبب اجساد قوي في فترة زمنية قصيرة
3. 100 min elapsed from the rupture of the membranes to the birth of calf where the amnion acid appears at the Vulva followed by the fore feet of calf then the head → trunk بعد 100 دقيقة من تمزق الغشاء المشيمي يظهر الامنيون في Vagina ثم الذراعين ثم الرأس ثم الجذع
4. with the birth of the anterior part of the calf, the cow may rises to a standing position & the remainder of the calf slips from the pelvis by gravity مع الولادة الجزء الامامي للبقرة، البقرة تقوم بتقف و الجبل هيبل من الموضع بالجاذبية
5. Parturition is longer in Cows that give birth to large calves & nervous heifers الولادة يتكون أطول للبقرة التي يولد بحول كبيرة أو عصبية

③ Post-Parturient Stage

The cow rests for available length of time, then gets up & licks the fetal membranes and fluids from Calves.
 لا يستطيع لم تمشي بلسانها اغشية الجنين والموائل هذا الوجه
The Placenta was eaten by 82% of the cattle

Maternal Behavior

Cows groom their calves during early Postpartum period Concentrating on the back & the abdomen

a - Bonding Formation:

- ① Contact between cow & her calf through 5 min postpartum in formation of strong Specific maternal Bond
- ② 50% the Calves will be rejected

التواصل بينهم خاضعون 5 دقائق من بعد الولادة
لوات اخرين 5 ما كان
وبعد 5 ساعة البقرة من تعرف بجلها هذا

b - Licking of fetal fluid:

- ① Stimulates the Subcutaneous blood & Lymph circulation of the calf ^{تعزيز الدورة الدموية واللمفية}
- ② Dryness of the calf & prevention of respiratory diseases ^{الجهاز التنفسي}
- ③ Increase muscular tone & stimulate first standing of calf after birth ^{اجفاف الجلد وحمايته من الامراض}
- ④ As the calf nurses, the cow will lick the Perineum stimulating urination & defecation by her calf & so getting rid of meconium ^{أول براز الرضيع}

Neonatal Behavior:

a) Coordinating recumbency:

Some calves will remain motionless for up to 30 min after Birth with ^{body} extension

⑤ Elevation: the young calves begin to stand with the

⑥ Ambulation: when the stance has been secured attempts at walking quickly initiated

لأنهم السريعة في الوقوف يحاول يمشي بسرعة

البحث عن الثدي والرضاعة

① Udder Searching and Nursing:

- * ① newborn calves begin to suckle with 2 to 5 hrs after birth
the neonate's first movement is towards his mom, it find extent with Random movement of his mother helping him to find it
- * ② they nuzzle & lick along the cow especially in high recess Axilla & Groin
Teats و Axilla & Groin
- * ③ Once the teat has been located for the 1st time the calf will be able to locate much more quickly & subsequent nursing
- * ④ New born calves normally nurse 5-7 times & lasts for 8-10 min
- * Best Calves 0-7 months old nurse for 3-5 times per day
- * ⑤ Typically calf stands alongside the mother facing caudally but sometimes calf's body will be at right angle to mother

Care of Cow after Parturition:

- ① Washing the external genitalia: - the flanks regions, ~~but back~~ with warm water & containing mild antiseptics as Crystal of Potassium Permanganate
- ② Parturient Cows should be given clean water & warm, to compensate the loss of great amount of fluids in calving and milk secretion
- ③ Cow should be given bran mash with warm water to provide a laxative effect to be easily digested food
- ④ Squeezing few streaks of milk from each quarter to remove any bacteria that might be present in the teats of udder

- 18م الحمل يكون أمه لمدة
أيام من أدوية الولادة مثل
أقوى الترابط بينهم
- the Cow & her Calf must be kept in the Calving Pen
- ② Breeding of the Cow = the post partum estrus comes during the first month after Calving → pregnancy 50% .
على جعل الحمل من (80-90) يوم بعد الولادة .

Care of calf at Calving time:

- ① if the Cow refuses to lick her Calf, Calf should be dried by straw or cloth
 - ② Remove fetal membrane
 - ③ navel cord should be cut
 - ④ Remove few streak milks from each quarter
 - ⑤ the Calf attempts to stand up in a few minutes
 - ⑥ Trimming the claws of the calf after birth to remove the gelatinous **Yellow** Horny Substance
 - ⑦ the Calf should always be given the **Colostrum** or first milk of dam
- الحمل السري لازم يقطع
وإن غفشت بكثيرا تدخل حصة البهيمن
هناك لون يصفى (بعضة أمهات) (بعضة أمهات)
قص عذائب الحمل لإزالة المادة الجيلاتينية الصفراء التي يتخلفها
أول لبن للأم

Management During Calf-Hood:

- Calf-Hood: Stage of life extends from birth to age of six months

General Management

Housing ① many Calves die from disease (to reduce this loss) → Cows should be Calved in a well-Cleaned and disinfected

- ② for the first several weeks the Calf should be kept in separate calf Pen

في الأسابيع الأولى للحمل لازم يتحاطر أوعية منفصلة (نظيفة ومنتظمة كويس) من الحظيرة و لا يخلط
و فاصل بينهم ارتفاع 1.0-1.15م

- ③ Racks allows the Calves to see each other and facilitate the free circulation of fresh air
- ④ While solid divisions lessen the risk of draughts and diseases

- ④ when Calves reach 3 months age they're best housed in small groups (3 m²/calf)

على يوصلوا مشهور يتحاطر و يهربون
و غيره و يكون المساحة 3م² لكل calf و لا يكونوا يتحاطر و يهربون 4م² لكل calf

و فاصل بينهم ارتفاع 1.0-1.15م

2. Feeding:

(a) Feeding Colostrum

البين الأول الأهم بقرنه من ٣ - ٥ أيام بعد الولادة ولازم يافده منها

* Colostrum contain 10% more protein & little more fat than normal milk

* Colostrum is normally rich Vit A, D and E

* Colostrum is rather to to Cuckaline ^{منه} & helps to clear intestines of young calf

* Colostrum is the most Supplier of protective antibodies to disease infections

Substitutes:

- (1) Colostrum from the mother (2) Frozen colostrum (3) Blood serum of cows (4) Egg white
Antibodies ^{gamma globulin} على أن يذوب في ماء ^{albumin}
amount of Colostrum (السلالة العادية) → (4.5 liter / day)

Natural Method:-

* This Method used mostly by farmers where the calf allowed sucking his mother directly

Sucking 8 to 10 min per feeding ^{أربعة}

Sucking 37 to 57 min per day ^{اليوم}

Calf at first 2 weeks → All the mother's udder milk ^{أيقول أنه يرضعها}

Calf at 3 weeks - 2 months → 1/2 the mother's udder milk ^{عشان يرضعها على}

Calf at 2 month to weaning → 1/4 the Mother's udder milk ^{الأكل مع اللبن}

• Weaning: ^{الإطعام}

The Calf weaned at (3.5 to 4 months) ^{يكون عنده ٣٥ إلى ٤ أشهر}

in farms weaning takes place when calf is weighed (80-90) Kg

Artificial Method:-

(a) Bucket Feeding: (1) When the Calf is about 4 days old normal & healthy

Take it away from the mother & teach it to drink whole milk from the pail

ما يكون عنده أيام طبيعي و ٤ أيام بناه و نعلمه يشرب اللبن من إناء من السفل

(2) If the Calf fails to get the idea at the first Trial, you may let it go hungry until the milking time

بمناسبة دجوع سوية بعد وجبة أو وجبتين وكذا لما نأكل نأكله تاني هيسر

(3) The amount of milk given to the Calf divided into two or three portions each day

B) Nipple Feeding

Nipple may be fitted on the pail and the calf allowed sucking this nipple
This Teat (هذا الثدي) لا يرضع به calf إلا أن يرضعها على الجريل ويضعها إلى

Advantages of Artificial feeding

- ① the actual yield of milk can be judged accurately
- ② the use of the calf for letting down of milk introduces an uncertain item as the calf may suck more milk or less
طالما سحبه الجبل كان اللبن ينزل فهو ينزل حسب وضاعف من سحبه الجبل لا يمكن أن يرضع أو أقل
- ③ the amount of milk which the calf should receive can be controlled.
- ④ when the calf dies, the cow will refuse to give milk, but nothing may happen in hand feeding